

ROBOTICS COURSE CONTENT

Duration- 6-8 Weeks

OBJECTIVE:

To provide learners with both **theoretical concepts** and **hands-on skills** in designing, building, and programming robots using sensors, actuators, microcontrollers, and control algorithms.

WEEK - 01

Introduction to Robotics

- History & evolution of robotics
- Types of robots: mobile, industrial, humanoid, autonomous
- Anatomy of a robot: sensors, actuators, controller, power, chassis
- Degrees of Freedom (DoF) and workspace
- Applications: manufacturing, defense, agriculture, healthcare

WEEK - 02

Electronics & Actuation Systems

- Basic electronics: Voltage, current, resistors, transistors
- Power supplies and batteries for robots
- Actuators: DC motors, servo motors, stepper motors
- Motor drivers (L298N, H-Bridge)
- Hands-on: Control a motor using Arduino or Raspberry Pi

WEEK - 02

Sensors & Feedback Systems

- Types of sensors: IR, Ultrasonic, LDR, Gyroscope, Accelerometer
- Sensor interfacing and calibration
- Concept of feedback in robotics
- PID control basics
- Practice: Line follower or obstacle-avoiding robot

WEEK - 04

Microcontroller Programming (Arduino/Pi)

- Introduction to Arduino and Raspberry Pi platforms
- Digital and analog I/O, PWM, interrupts
- Serial communication (UART, I2C, SPI)
- Writing & uploading Arduino code
- Hands-on: Build a basic robotic arm or light tracker

WEEK - 05

Mobile Robotics & Navigation

- Types of mobile robots (wheeled, tracked, legged)
- Kinematics of differential drive robot
- Path planning basics
- Localization and mapping (Intro to SLAM)
- Practice: Autonomous navigation using sensors

WEEK - 06

Robotics Control Systems

- Open-loop vs closed-loop control
- PID tuning for motor control
- Introduction to inverse kinematics (for robotic arms)
- MATLAB/Simulink (optional) for simulating control systems

OPTIONAL

WEEK - 07

IoT, Wireless & AI in Robotics

- IoT integration for remote robot control
- Bluetooth/Wi-Fi modules (HC-05, ESP32)
- Voice-controlled robots using Google Assistant
- Object tracking using OpenCV + Python
- Hands-on: Mobile app-controlled robot

WEEK - 08

Capstone Project & Certification

- Capstone Project Ideas:
 - Pick and Place Robotic Arm
 - Firefighting Robot
 - Maze Solver Robot
 - Surveillance Drone on Wheels
 - Final Report & Demonstration
 - Evaluation & Certification

Hardware & Tools Used

- Controllers: Arduino Uno/Nano, Raspberry Pi, ESP32
- Actuators: Servo motors, DC motors, Stepper motors
- Sensors: IR, Ultrasonic, LDR, DHT11, MPU6050
- Software: Arduino IDE, Python, Proteus/Tinkercad (for simulation), MATLAB (optional)

Course Deliverables

- Code files and libraries
- Circuit diagrams and datasheets
- Weekly quizzes and lab tasks
- Capstone project report format
- Certificate of Completion